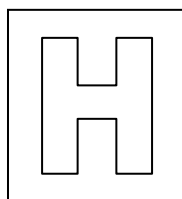


Candidate Name: _____

Class Adm No

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2011 Preliminary Examination II

Pre-university 3

H2 Biology

9648/01

23 September 2011

1 hour 15min

Additional Materials: Multiple Choice Answer Sheet

READ THESE INSTRUCTIONS FIRST

Write in soft pencil.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Write your name, Centre number and index number on the Answer Sheet in the spaces provided unless this has been done for you.

There are **forty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A, B, C** and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

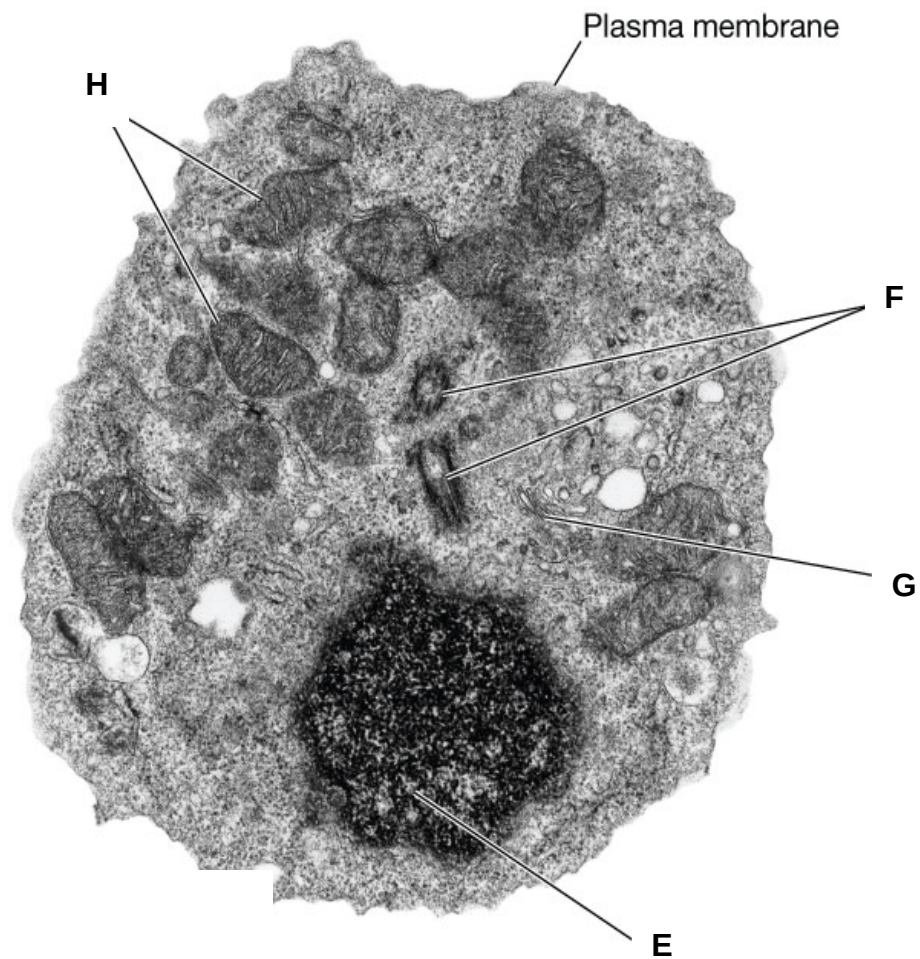
Each correct answer will score one mark. A mark will not be deducted for a wrong answer. Any rough working should be done in this booklet.

Calculators may be used.

This question paper consists of 20 printed pages.

[Turn over

1. An electron micrograph of a cell is shown below.



Match the organelles **E, F, G and H** associated with the cellular processes listed.

	E	F	G	H
A	DNA replication	Digestion of material	Organizes the spindle	Oxidative phosphorylation
B	Oxidative phosphorylation	Organizes the spindle	Digestion of material	DNA replication
C	Organizes the spindle	Digestion of material	Oxidative phosphorylation	Packaging of secretory products
D	DNA replication	Organizes the spindle	Packaging of secretory products	Oxidative phosphorylation

2. At extremely low oxygen levels, important cellular proteins unravel. Another intracellular protein, Hsp60, has been discovered in a wide range of prokaryotic and eukaryotic cells. This protein may bind to other proteins and prevent their unraveling.

It would be reasonable to infer that Hsp60

- A. is manufactured at free ribosomes.
 - B. has a structure which is secondary.
 - C. is produced continuously by a cell.
 - D. is produced in equal amounts by living cells in a multicellular organism.
3. Particular biological molecules react with chemicals called reagents to give distinct colour changes. The colour depends on the kind of biological molecule and the type of reagent used, as shown in the following table.

Chemical reagent	Biological molecule	Colour change observed
K	starch	dark blue
L	protein	violet
M	lipid	red
N	nucleic acid	green

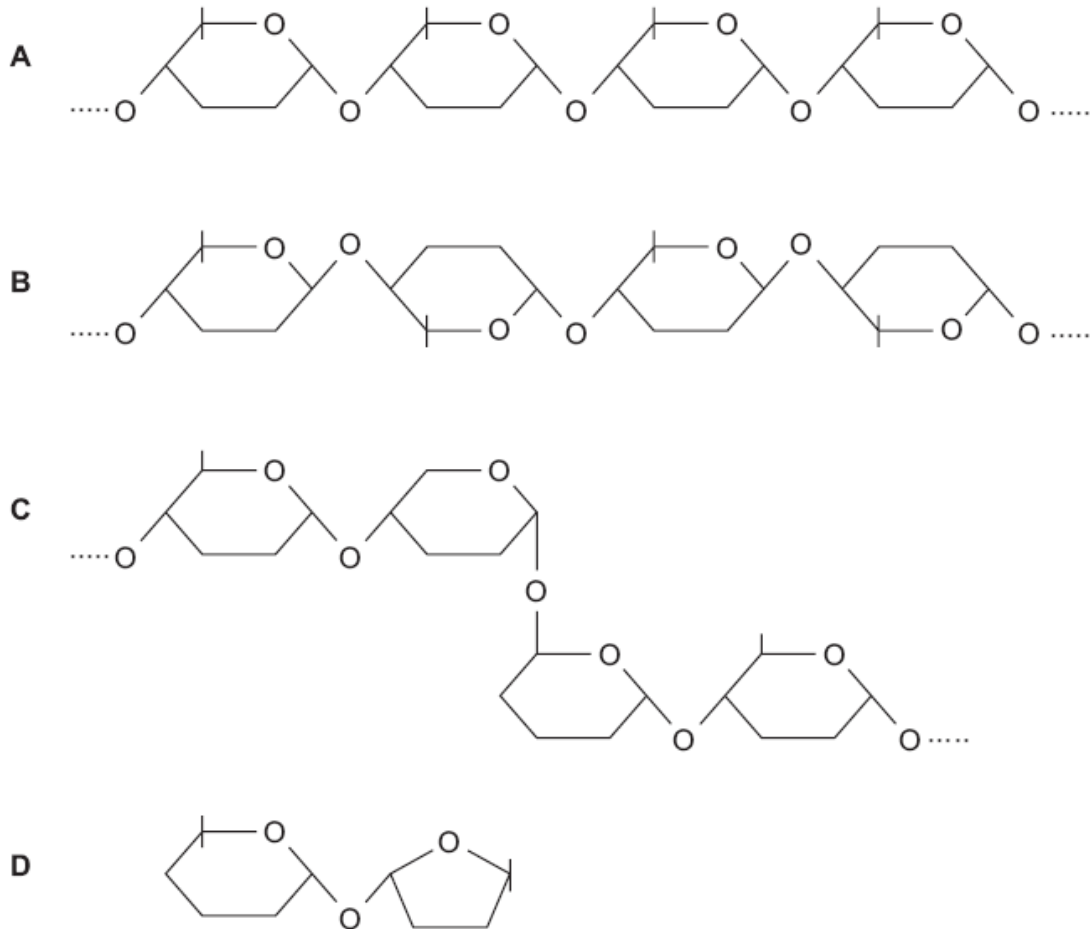
A researcher added different reagents to some isolated ribosomes. The colours she would observe are

- A. red and green.
 - B. green and violet.
 - C. dark blue and red.
 - D. violet and dark blue.
4. Mitochondria are thought to have evolved from prokaryotic cells that were ingested by an ancestral cell.

Which feature have the prokaryotes lost during their evolution into mitochondria?

- A. cell wall
- B. circular chromosome
- C. endoplasmic reticulum
- D. ribosomes

5. Which diagram shows part of a structural polysaccharide?

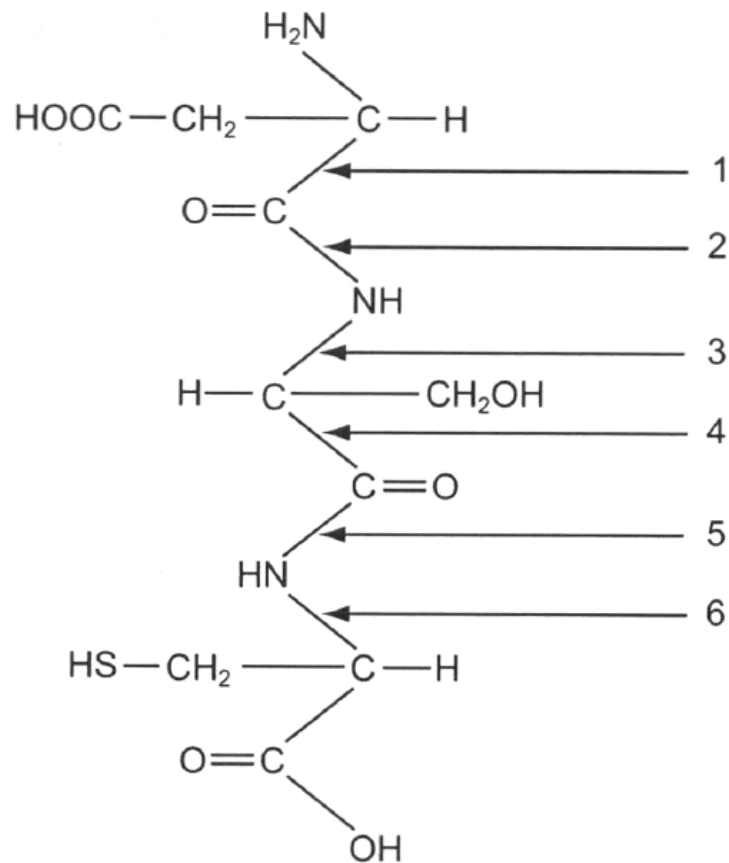


6. Some foods contain 'hydrogenated vegetable oils'. These are unsaturated fats that have been converted to saturated fats.

Which property of the fats will have changed?

- A.** Their hydrocarbon chains will fit together more closely.
- B.** Their solubility in water will increase.
- C.** They will have more double bonds in their molecules.
- D.** They will remain liquid at room temperature.

7. The diagram represents a tripeptide.



At which bonds does hydrolysis occur to release the amino acids from the tripeptide?

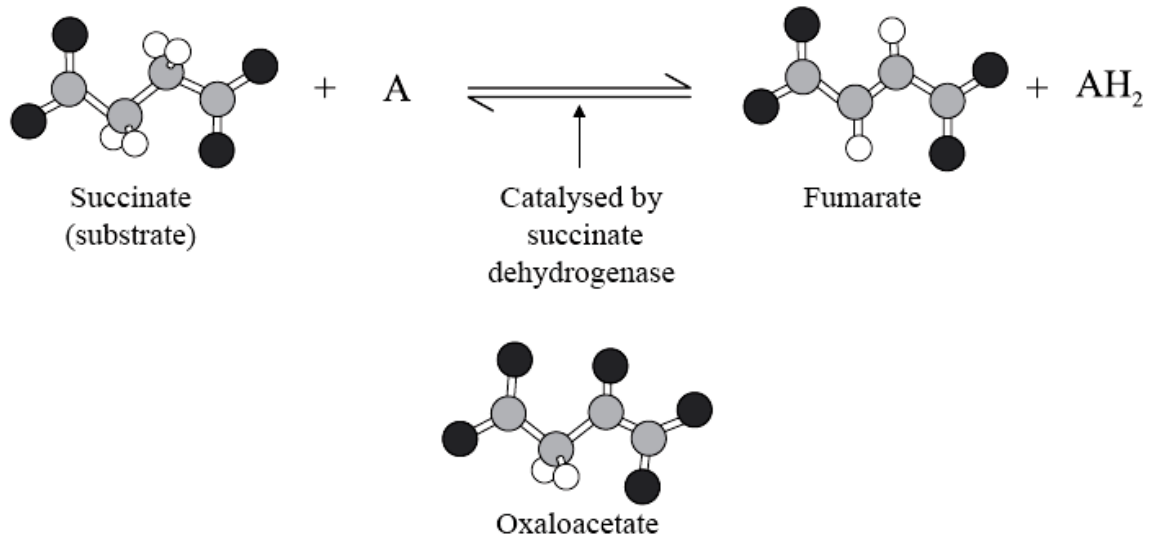
- A. 1 and 4
 B. 2 and 5
 C. 2 and 6
 D. 3 and 6
8. Which levels of protein structure can determine the specificity of an enzyme?
- 1 primary
 2 secondary
 3 tertiary
 4 quaternary
- A. 1, 2, 3 and 4
 B. 1, 2 and 3 only
 C. 1, 2 and 4 only
 D. 3 and 4 only

9. Laundry powder is sometimes advertised as containing powerful enzymes that break down dirt. These enzymes are called extremozymes. They come from some species of bacteria and archaea. The following table gives the optimal functioning of enzymes from some of these species.

Species	Enzyme	Optimal temperature °C	Optimal pH
<i>Psychrobacter</i> sp.	J	10–30	7.0–9.0
<i>Pseudomonas</i> sp.	K	40	10.0
<i>Methanococcus</i> sp.	L	120	5.0–8.0
<i>Cystofilobasidium</i> sp.	M	40–42	5.0

Given this information and your knowledge of enzyme function, the best enzyme to add to the laundry powder would be

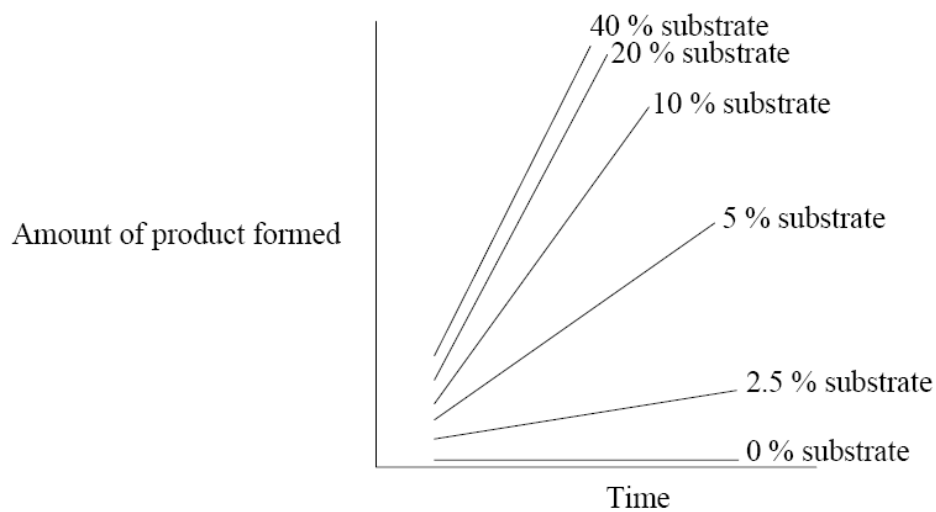
- A. enzyme J.
 - B. enzyme K.
 - C. enzyme L.
 - D. enzyme M.
10. Why is oxaloacetate a competitive inhibitor?



[Source: image from WK Purves, *et al.*, (2003) *Life: The Science of Biology*, 4, Sinauer Associates (www.sinauer.com) and WH Freeman (www.whfreeman.com)]

- A. It causes a conformational change to the active site.
- B. It binds to the enzyme away from the active site.
- C. It is structurally similar to succinate.
- D. It is structurally similar to succinate dehydrogenase.

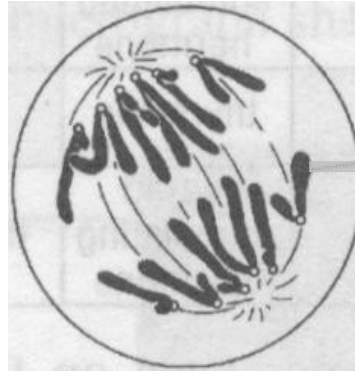
11. The graph below shows the effect of changing the substrate concentration on an enzyme controlled reaction



What is the correct interpretation of these data?

- A. The rate of reaction increases continuously with increase in substrate concentration.
- B. The rate of reaction decreases continuously with increase in substrate concentration.
- C. The rate of reaction increases up to a point and then remains constant.
- D. The rate of reaction is not affected by any change in the substrate concentration.

12. The figure below shows a diploid plant cell undergoing anaphase.



Taxanes which inhibit spindle fibres formation are added to another cell from the same plant during anaphase II. If the amount of DNA in the diagram above is x , which one of the following could be one possible result of a gamete formed?

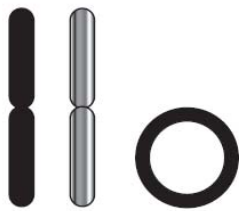
	number of chromosomes	amount of DNA
A	6	$\frac{1}{2} x$
B	3	$\frac{1}{2} x$
C	6	$\frac{1}{4} x$
D	3	$\frac{1}{4} x$

13. The following diagram shows chromosome pair 3 and the mitochondrial DNA from a skin cell of a female and a male.

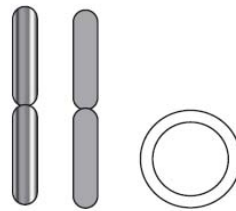


In relation to the chromosome pair shown and mitochondrial DNA, the genetic makeup of an offspring from this couple could be

A.



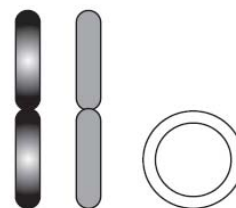
B.



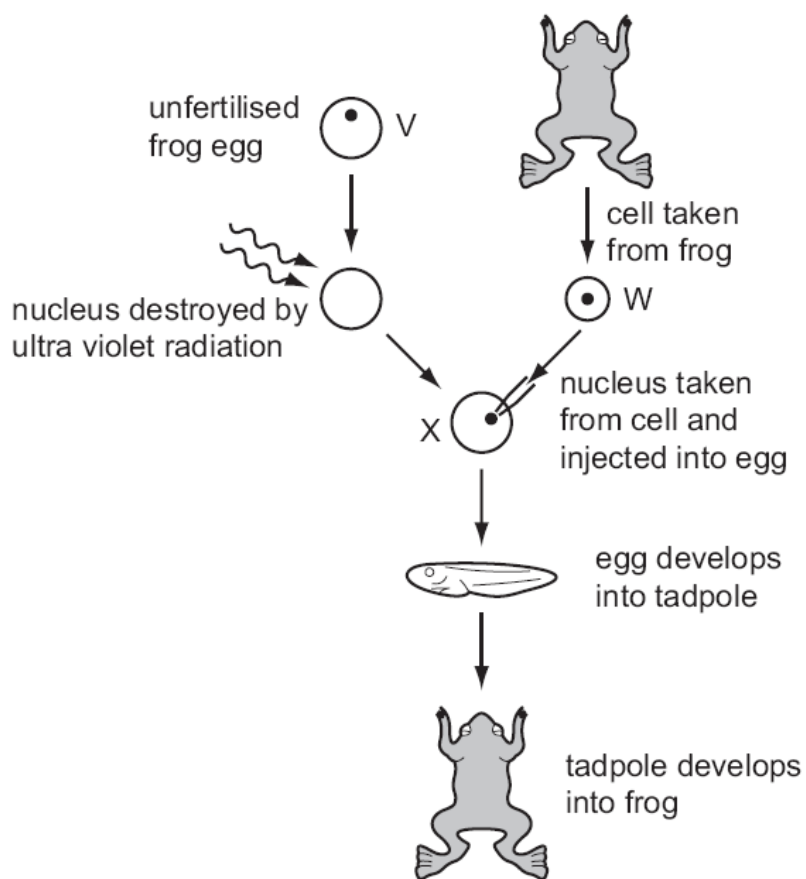
C.



D.



14. The diagram shows how genetically identical frogs can be developed from unfertilised frog eggs. The diploid number ($2n$) for frogs is 26.



Which combination of numbers correctly identifies the number of chromosomes in each of the types of cell in the diagram?

	V	W	X
A	13	13	26
B	13	26	13
C	13	26	26
D	26	26	13

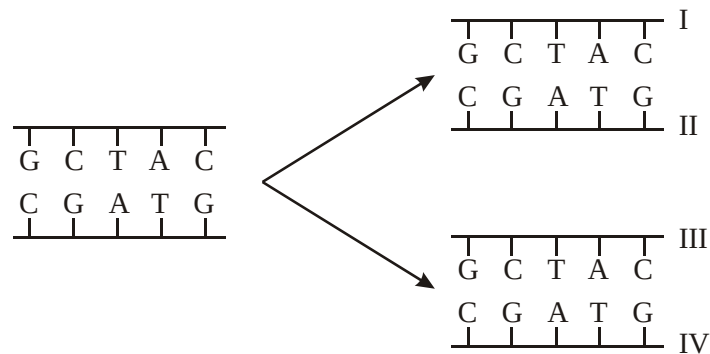
15. The following events occur during transcription.

- 1 Bonds break between complementary bases
- 2 Bonds form between complementary bases
- 3 Sugar-phosphate bonds form
- 4 Free nucleotides pair with complementary nucleotides

Before the mRNA leaves the nucleus, which events will have occurred twice?

- A. 1 and 2 only
- B. 1, 3 and 4 only
- C. 2, 3 and 4 only
- D. 1, 2, 3 and 4

16. The diagram below shows a short section of DNA molecule before and after replication. If the nucleotides used to replicate the DNA were radioactive, which strands in the replicated molecules would be radioactive?



- A. II and III only
- B. I and III only
- C. I and II only
- D. I, II, III and IV

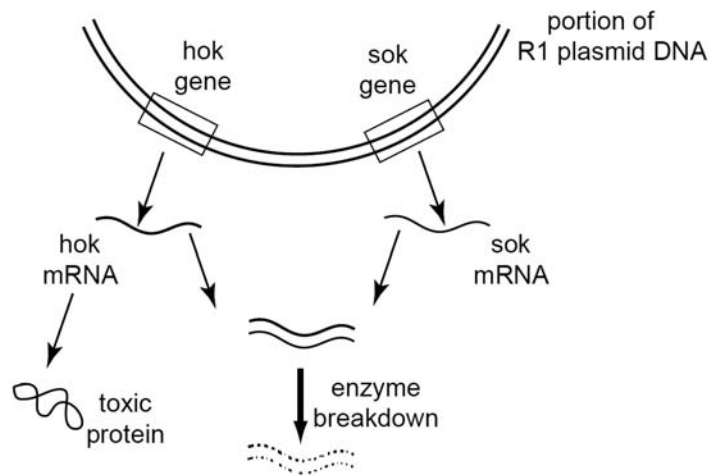
17. Why are viruses referred to as obligate parasites?

- A. They cannot reproduce outside of a host cell.
- B. Viral DNA always inserts itself into host DNA.
- C. They invariably kill any cell they infect.
- D. They must use enzymes encoded by the virus itself.

18. Viral genomes vary greatly in size and may include from four genes to several hundred genes. Which of the following viral features is most apt to correlate with the size of the genome?
- A. size of the viral capsomeres
 - B. RNA versus DNA genome
 - C. double versus single strand genomes
 - D. size and shape of the capsid
19. Which of the following statements describes the lysogenic cycle of lambda (λ) phage?
- A. After infection, the viral genes immediately turn the host cell into a lambda-producing factory, and the host cell then lyses.
 - B. Most of the prophage genes are activated by the product of a particular prophage gene.
 - C. The phage genome replicates along with the host genome.
 - D. Certain environmental triggers can cause the phage to exit the host genome, switching from the lytic to the lysogenic.
20. To cause a human pandemic, the H5N1 avian flu virus would have to
- A. spread to primates such as chimpanzees.
 - B. develop into a virus with a different host range.
 - C. become capable of human-to-human transmission.
 - D. arise independently in chickens in North and South America.
21. Regarding prokaryotic reproduction, which statement is **correct**?
- A. Prokaryotes form gametes by meiosis.
 - B. Prokaryotes feature the union of haploid gametes, as do eukaryotes.
 - C. Prokaryotes exchange some of their genes by conjugation, the union of haploid gametes, and transduction.
 - D. Mutation is a primary source of variation in prokaryote populations.

22. The role of a metabolite that controls a repressible operon is to
- A. bind to the promoter region and decrease the affinity of RNA polymerase for the promoter.
 - B. bind to the operator region and block the attachment of RNA polymerase to the promoter.
 - C. bind to the repressor protein and inactivate it.
 - D. bind to the repressor protein and activate it.
23. A researcher found a method she could use to manipulate and quantify phosphorylation and methylation in embryonic cells in culture. In one set of experiments she succeeded in decreasing methylation of histone tails. Which of the following results would she most likely see?
- A. increased chromatin condensation
 - B. decreased chromatin concentration
 - C. abnormalities of mouse embryos
 - D. decreased binding of transcription factors
24. Transcription factors in eukaryotes usually have DNA binding domains as well as other domains also specific for binding. In general, which of the following would you expect many of them to be able to bind?
- A. repressors
 - B. ATP
 - C. protein-based hormones
 - D. other transcription factors

25. The *hok* gene in the R1 plasmid of the bacterium *E. Coli* codes for a toxic protein that causes cell death. The *sok* gene in the same plasmid codes for an antisense regulator that binds to the *hok* mRNA. When the two mRNA strands come together, they are broken down by an enzyme and so cell death is prevented. The following diagram summarises these events.



It would be reasonable to expect that

- A. *hok* mRNA would be translated if *sok* mRNA was absent.
 - B. *hok* DNA would be unable to be transcribed if *sok* mRNA is present.
 - C. *sok* mRNA would have the same nucleotide sequence as *hok* mRNA.
 - D. an *E. Coli* cell without any plasmids would be killed by the toxic protein.
26. In some people, hemoglobin always contains the amino acid valine in place of a glutamic acid at one position in the protein. What is the cause of this?
- A. An error in transcription of the hemoglobin gene
 - B. An error in translation of the mRNA
 - C. Lack of glutamic acid in the diet
 - D. A base substitution in the hemoglobin gene

27. Some people are genetically predisposed to suffer from particular diseases. The table shows some diseases and their causes.

	disease	cause
1	Alzheimer's syndrome	myeloid protein plaques in the brain
2	brittle bone disease	abnormal collagen
3	Down's syndrome	extra chromosome 21
4	sickle cell anaemia	abnormal haemoglobin
5	Turner's syndrome	absence of a second sex chromosome
6	Cri-du-chat syndrome	missing part of chromosome 5

Which diseases are caused by gene mutation and which by chromosome aberration?

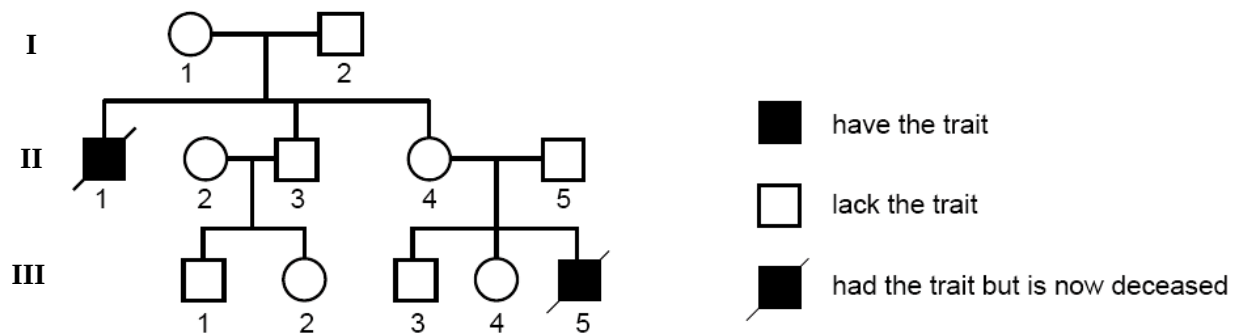
	gene mutation	chromosome aberration
A	1 and 3	2 and 5
B	2 and 4	3 and 6
C	3 and 5	2 and 4
D	4 and 6	3 and 5

28. One hereditary disease in humans, called xeroderma pigmentosum (XP), makes homozygous individuals exceptionally susceptible to UV-induced mutation damage in the cells of exposed tissue, especially skin. Without extraordinary avoidance of sunlight exposure, patients soon succumb to numerous skin cancers. Which of the following best describes this phenomenon?

- A.** inherited cancer taking a few years to be expressed
- B.** embryonic or fetal cancer
- C.** inherited predisposition to mutation
- D.** inherited inability to repair UV-induced mutation

29. What is a genetic test cross?
- Testing a suspected homozygote by crossing it with a known heterozygote
 - Testing a suspected heterozygote by crossing it with a known heterozygote
 - Testing a suspected homozygote by crossing it with a known homozygous dominant
 - Testing a suspected heterozygote by crossing it with a known homozygous recessive
30. Which of the following statements describes proto-oncogenes?
- They can code for proteins associated with cell growth.
 - They are introduced to a cell initially by retroviruses.
 - They are produced by somatic mutations induced by carcinogenic substances.
 - Their normal function is to suppress tumor growth
31. Duchenne Muscular Dystrophy (DMD) is inherited as an X-linked recessive condition in which cells fail to produce normal dystrophin protein.

The following pedigree shows a family in which some members have DMD.



From the pedigree it is reasonable to conclude that

- II 4 is homozygous normal at the DMD locus.
- I 1 is heterozygous with respect to the DMD allele.
- I 1 and I 2 have a one-in-four chance of producing a daughter with DMD.
- II 4 and II 5 have a one-in-three chance of producing a daughter who is a carrier of DMD.

32. Blood samples taken from an individual who had been fasting for 24 hours would have which of the following?

- A.** high levels of insulin, high levels of glucagon
- B.** low levels of insulin, high levels of glucagon
- C.** low levels of insulin, low levels of glucagon
- D.** high levels of insulin, low levels of glucagon

33. Protons are pumped into the intermembrane space in the mitochondrion during electron transport. These protons diffuse through proteins to be used in the synthesis of ATP.

Why do these protons diffuse only through the proteins in the membrane?

- A.** They are repelled by the charge on the phospholipids.
- B.** They are too large to pass between the phospholipids.
- C.** They are water soluble and cannot pass through the phospholipids.
- D.** They attach to the polar heads of the phospholipids.

34. The free energy for the oxidation of glucose to CO_2 and water is -686 kcal/mol and the free energy for the reduction of NAD^+ to NADH is $+53 \text{ kcal/mol}$.

Why are only two molecules of NADH formed during glycolysis when it appears that as many as a dozen could be formed?

- A.** Most of the free energy available from the oxidation of glucose is used in the production of ATP in glycolysis.
- B.** Glycolysis is a very inefficient reaction, with much of the energy of glucose released as heat.
- C.** Most of the free energy available from the oxidation of glucose remains in pyruvate, one of the products of glycolysis.
- D.** There is no CO_2 or water produced as products of glycolysis.

35. What is the mechanism of natural selection?

- A.** Any individuals in a population can be selected entirely by chance.
- B.** After a change in the environment a species will evolve adaptations to the new conditions.
- C.** If an adaptation to the environment is useful, an individual will develop it and pass it on to its offspring.
- D.** Variations amongst individuals of a population are selected by a changing environment.

36. Which of the following are used as evidence for evolution?

- I. Homologous structures
- II. Selective breeding of domesticated animals
- III. Overproduction of offspring

- A. I and II only
- B. I and III only
- C. II and III only
- D. I, II and III

37. A student performed the following steps in a Southern blot experiment to determine the number of copies of a particular gene that has been inserted in a genetically modified organism.

1. Transfer of DNA to nitrocellulose membrane.
2. Restriction digestion of genomic DNA.
3. Cleaved DNA separated using gel electrophoresis.
4. Create radioactive probe.
5. Incubate probe and membrane.

Which is the correct sequence to the above steps?

- A. 2 → 3 → 1 → 4 → 5
- B. 2 → 3 → 1 → 5 → 4
- C. 4 → 5 → 1 → 2 → 3
- D. 5 → 4 → 3 → 2 → 1

38. Which of the following is **not** a social or ethical consideration of Human genome project?

- A. detect genetic predispositions to disease
- B. fairness in accessibility of data
- C. reliability of genetic tests for pre-natal decision making
- D. employers to demand DNA testing

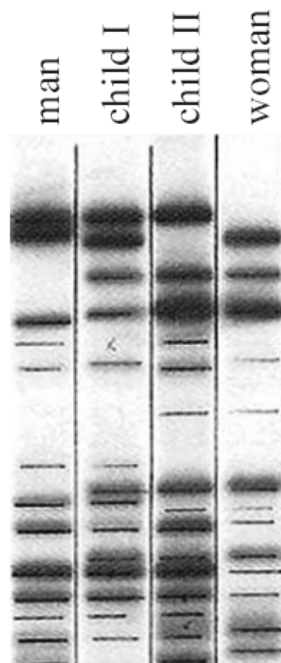
39. One type of genetically modified corn has

- A gene for the production of Bt toxin which protects the plant against a specific insect;
- An 'amp'-gene which was introduced in the plant together with the Bt toxin gene. This gene gives resistance to the antibiotic ampicillin.

There is concern that the 'amp' gene may transfer to enterobacteria in the human intestine during nucleic acid digestion making treatment with ampicillin ineffective for diseases caused by enterobacteria.

Which statement explains why the transfer of this gene from the plant to bacteria in the human intestine is unlikely?

- An origin of replication and appropriate prokaryotic promoters are required for the 'amp' gene to be expressed.
 - Bacteria took up DNA released during digestion of the plants by human nuclease enzymes, without the presence of a vector.
 - 50% of the enterobacteria isolated from humans are 'amp' resistant.
 - All plant DNA is digested and destroyed in the human intestine during digestion of plant cells by human nuclease enzymes.
40. What conclusion can be made from the following evidence from an analysis of DNA fragments?



- Both children are related to both parents.
- Child I is related to the man but child II is not.
- Both children are unrelated to either of the parents.
- Child II is related to the man but child I is not.

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